AI Data Centers and Capacitor Availability:

Murata's Capacitor Book-to-Bill vs. Nvidia Data-Center Revenue, 2024–2026

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October 2026

1 Motivation

Multilayer ceramic capacitors (MLCCs) are tiny components that stabilize the power supply to chips. An AI server board uses thousands of them; industry estimates put AI servers at 10–15 times the MLCC content of a general-purpose server, and they rely on high-capacitance and high-voltage parts. This project asks whether the AI data-center buildout has tightened the *availability* of capacitors, and whether availability tightened before prices responded.

In a supply-constrained market, a demand shock can show up first as longer waits and growing order backlogs rather than as higher prices, because large buyers purchase on contracts that reset with a lag. Measuring availability directly is therefore central to the analysis.

2 Data

- **Murata Manufacturing**, the largest MLCC producer: quarterly orders received, sales shipped and order backlog for the capacitor segment (millions of yen), from Murata's quarterly *Consolidated Financial Results* (July 2025 – July 2026 reports). January–March quarters are derived as full-year minus nine-month totals.
- **Nvidia**: quarterly data-center revenue ($ billions) from Nvidia's quarterly results releases. Some quarters are derived from reported growth rates. Nvidia's fiscal quarters end about one month after Murata's, so each Murata quarter is paired with the Nvidia quarter that mostly overlaps it.

3 Measure of availability

The outcome is Murata's capacitor *book-to-bill* ratio:

$$\mathrm{BTB}_t = \frac{\mathrm{Orders}_t}{\mathrm{Sales}_t}. \tag{1}$$

A value above 1 means customers are ordering more than the company can ship, so unfilled orders accumulate and availability tightens.

Quarter	Orders (JPY mn)	Sales (JPY mn)	Book-to-bill	Nvidia DC ($B)
Apr–Jun 2024	213,226	203,313	1.05	26.3
Jul–Sep 2024	204,681	213,999	0.96	30.7
Oct–Dec 2024	207,210	213,089	0.97	35.6
Jan–Mar 2025	207,567	201,444	1.03	39.3
Apr–Jun 2025	223,995	217,330	1.03	41.0
Jul–Sep 2025	241,807	237,572	1.02	51.1
Oct–Dec 2025	268,055	239,114	1.12	62.3
Jan–Mar 2026	329,712	242,402	1.36	75.4
Apr–Jun 2026	415,505	282,508	1.47	89.0

Table 1: Murata capacitor segment and paired Nvidia data-center revenue.

4 Method

I report the Pearson correlation, the Spearman rank correlation, and an ordinary least squares regression:

$$\mathrm{BTB}_t = \alpha + \beta\,\mathrm{NvidiaDC}_t + \varepsilon_t, \tag{2}$$

where $\mathrm{NvidiaDC}_t$ is Nvidia's data-center revenue in the overlapping quarter. The analysis is implemented in base R (`murata_vs_nvidia_base.R`).

5 Results

Statistic	Value
Pearson r	0.92
Spearman ρ	0.70
$\hat{\beta}$ (per $1B of Nvidia DC revenue)	0.0078 (s.e. 0.0012)
$\hat{\alpha}$	0.723 (s.e. 0.067)
t-statistic on $\hat{\beta}$	6.28
p-value on $\hat{\beta}$	0.0004
R^2	0.85
Observations	9

Table 2: Book-to-bill regressed on Nvidia data-center revenue.

Each additional $10 billion of quarterly Nvidia data-center revenue is associated with roughly a 0.08 increase in Murata's capacitor book-to-bill. Through September 2025, orders and shipments were balanced (book-to-bill ≈ 1) as Murata expanded output. Once Nvidia's quarterly growth re-accelerated in late 2025, orders pulled ahead: book-to-bill rose to 1.12, 1.36 and 1.47 over the next three quarters, and Murata's backlog rose from about 2 to 4.3 months of sales. The July–September 2025 quarter sits below the fitted line, consistent with a one-quarter lag.

Over the same period Murata reported falling selling prices in every quarter through June 2026, and the U.S. capacitor producer price index rose only about 6–8% per year, so availability tightened before prices responded. Industry reports from mid-2026 describe the first broad MLCC price increases and lead times of 26–40 weeks for high-end parts.

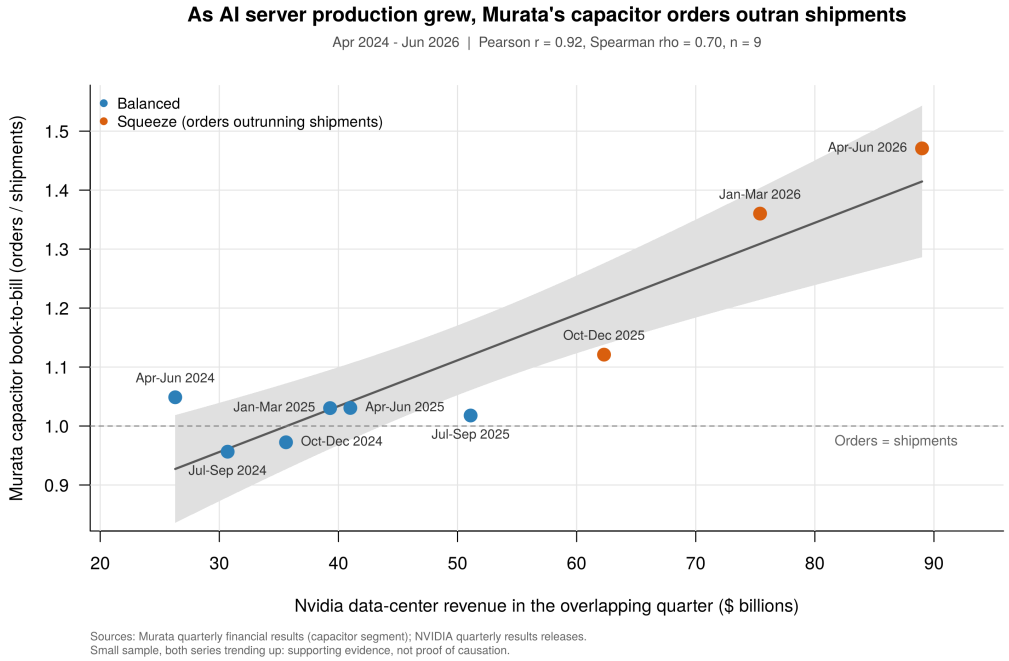


Figure 1: Murata capacitor book-to-bill vs. Nvidia data-center revenue by quarter.

6 Limitations

- The sample is small (nine quarters) and both series trend upward, so the correlation overstates precision; the results are evidence consistent with AI demand tightening capacitor availability, not proof of causation.
- Murata's figures are in yen; the yen weakened about 10% over the period, which inflates yen growth rates.
- Other MLCC demand (automotive, smartphones) and possible double ordering by customers may contribute to the rise in orders.

7 Sources

- Murata Manufacturing Co., Ltd., *Consolidated Financial Results* (quarterly), `https://corporate.murata.com/en-global/ir`.
- NVIDIA Corporation, quarterly financial results press releases, `https://investor.nvidia.com`.
- TrendForce, "MLCC lead times diverge as high-end products stretch to 5–10 months" (Aug. 2026).
- U.S. Bureau of Labor Statistics, PPI: Capacitors for Electronic Circuitry (WPU11781148), via FRED.